Joint Tracking of Moving Target in Single-Channel Video SAR

Chao Zhong, Jinshan Ding, *Member, IEEE*, and Yuhong Zhang, *Senior Member, IEEE*

***Abstract*— Video synthetic aperture radar (SAR) has been found very useful in ground moving target indication (GMTI) and tracking. The dynamic shadows in video SAR imagery sequences indicate the real positions of moving targets, which can be utilized in target detection and tracking. Unfortunately, the shadow-based method often fails when the shadows are not sufficiently developed. On the other hand, the traditional energy-based GMTI methods exhibit performance degradation when SAR images of a moving target are distorted or smeared. Neither of these two methods can stand alone to provide robust detection and tracking of moving targets. This article presents a joint processing framework for video SAR GMTI and tracking by combining the target shadow and echo energy information, which effectively lowers the false alarm and miss-detection rate. This approach is very suitable for real-time surveillance of ground moving targets in a single-channel video SAR system. The proposed approach has been verified by using the simulated and the real video SAR data.**

***Index Terms*— Ground moving target indication (GMTI), moving target tracking, radar imaging, shadow detection, video synthetic aperture radar (SAR).**

I. Introduction

UNLIKE conventional synthetic aperture radar (SAR) that provides stationary radar imagery, video SAR continuously produces imagery of the illuminated scene at a frame rate [1]–[5]. Video SAR has the potential to obtain useful data, such as positions, velocities, and trajectories of moving targets in real time, which are highly desired in many SAR applications.

In classical SAR ground moving target indication (GMTI), a moving target is detected through its echo energy against the ground clutter. The typical single-channel techniques, such as the frequency domain filtering method [6] and time-frequency analysis method, [7], [8], and the multichannel techniques, such as the displaced phase center antenna (DPCA) [9], [10] and along-track interferometry (ATI) [11], [12], can be used to find moving targets. Page *et al.* [13] defined an approach for detecting and tracking moving vehicles by mitigating the combined effects of moving target defocus and strong clutter by multichannel SAR. The detection and tracking performance of these methods depends on the output signal to clutter and noise ratio (SCNR). Although the multichannel techniques usually perform better because of their strong capability in clutter suppression, the single-channel techniques are often preferred for the sake of system simplicity and low cost. Poisson *et al.* [14] reconstructed the trajectories of moving targets from the apparent coordinates by inverting an equation system of measurements in a single-channel SAR. The precision of the reconstruction relies on the constant velocity hypothesis and the angular span, and it is hard to realize the target tracking in real-time. Henke *et al.* [15], [16] proposed a framework by combining exo- and endo-clutter processing for tracking moving targets with prior information about the road network for a more accurate tracking. A roadmap-aided method is also proposed in [17] for reconstructing the trajectory in a single-channel circular SAR, where a dynamic programming (DP) algorithm is proposed to project the range-Doppler (RD) domain trajectories on the known road grid. These methods are applicable when the imaging region is well known *a priori* and are suffered from the minimum detectable velocity (MDV). It is difficult for a single-channel system to locate moving targets accurately by utilizing the position-shifted and defocused echo energy. Moreover, the echo energy of a moving target can be completely defocused/diffused when the target is highly maneuvering, which severely degrades the detection performance.

Fortunately, a moving target could leave a sequence of dynamic shadows at its real locations in the video SAR imagery when the target images are position-shifted and/or defocused. The dynamic shadows between adjacent frames provide a real track of the moving target and are independent of the target radar cross section (RCS). The shadow-based methods for detecting and tracking moving targets in video SAR have received much attention recently.

Representative shadow detection approaches based on classical image processing can be found in [18] and [19]. In [20], the moving targets in SAR imagery are detected by tracking the shadow over multiple looks. In [21], we presented a framework for shadow-aided moving target detection using deep neural networks (DNNs) in video SAR. Tian *et al.* [22] suggested an expanding and shrinking strategy-based track-before-detect (ES-TBD) algorithm to detect and track moving targets in video SAR sequential images. Zhao *et al.* [23] utilized the background-aware correlation filter (BACF) and

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proposed a saliency-based detection mechanism to track shadows in video SAR. However, the shadow of a moving target may not be held steady due to many time-variant factors such as the moving velocity, direction, acceleration, and imaging geometry. The detection of target shadows is conditioned on the background and noise level as well. It is difficult to distinguish a target shadow from weak background in an SAR image, especially when the shadow is quite noisy [24], [25]. A slow-moving target could be partially focused, so that the target shadow may be contaminated by its own image. When a higher-speed target passes through a background area during the SAR integration time, partial blockage of the moving target on the background results in an indistinctive shadow. Besides, the position-shifted image of some other moving target may occur in the shadow area formed by the moving target of interest. The sidelobes of strong scattering objects could also smear the shadow area. All these issues may result in unreliable detection and tracking, especially in an environment involving multiple maneuvering targets.

Consequently, neither an energy-based method nor a shadow-based method can individually provide a robust detection and tracking of moving targets. One feasible solution is to combine the target echo energy and the shadow information to improve the detection performance. Xu *et al.* [26] proposed an enhanced shadow-aided detection approach for a multichannel SAR-GMTI system where a knowledge-aided shadow detection algorithm with an adaptive threshold is used to mitigate the influence of target motion. However, undeveloped target shadow, which often happens for a maneuvering target, will lead to detection failure. In [27], the radar data is processed with various parallel combinations of video SAR mode, exo-clutter mode, and clutter attenuated Doppler mode, so that the processing can be "tuned" to a proper target velocity. The corresponding video result [28] released by the Sandia National Laboratory shows that a moving target can be tracked effectively with the parallel processing.

This article presents the joint framework for robust moving target tracking in video SAR. Distinguished from [27] where the moving target tracking is gone through with tuning the processing to a proper mode, this article combines the target shadow and echo energy information between frames in video SAR. This joint processing makes the target tracking more robust and can adapt to the targets in different motion states, even stopped targets. Specifically, we make the shadow-based and energy-based detections first at a high probability of detection, separately, and then perform interframe data association to get the candidate tracks preliminarily. False tracks will be eliminated by associating the shadow-detected tracks in imagery with the energy-detected tracks in RD domain. Track update and management methods are also proposed to handle different cases of moving targets. The experimental results show that the developed framework can provide a satisfactory moving target tracking performance in video SAR.

This article is organized as follows. Section II briefly discusses the features of moving target in video SAR. Section III describes the proposed framework for tracking moving targets in video SAR in detail. The effectiveness of the proposed approach is verified and demonstrated by simulated and real video SAR data in Section IV. Section V summarizes this article.

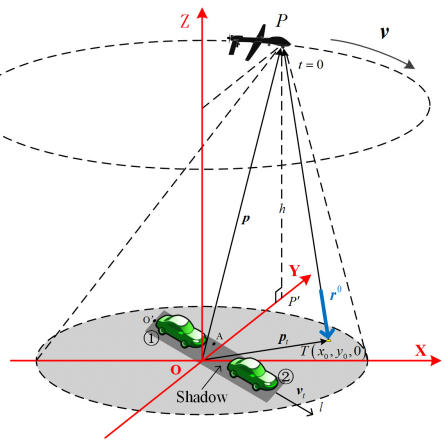

Fig. 1. Geometry and Cartesian coordinate systems for video SAR imaging in circular spotlight mode.

II. Features of Moving Target in Video SAR

The SAR images of a moving target are often shifted and smeared along the direction of cross-range with the change of observation angle, which leads to a shadow left at its real position in the imagery. Video SAR is expected to produce imagery at a required frame rate. The short integration time and high carrier frequency featured by a typical video SAR system introduce some new characteristics of moving targets in the imagery.

Consider a video SAR that works in circular spotlight mode to monitor a scene of interest continuously, as shown in Fig. 1 for an illustrative geometry and the Cartesian coordinate system with the origin O being at the center of the imaging scene, where the radar platform moves at a constant height of h with a constant velocity of υ. The radar instant location is at $P(R_d\sin(\omega t), R_d\cos(\omega t), h)$, where R_d denotes the distance from the nadir P' to the center of the scene O, ω is the angular velocity of the platform, and t denotes the slow time. The frame rate can be expressed as

$$fps = \frac{1}{(1-\gamma)T_a} = \frac{2\rho_a f_c \upsilon}{(1-\gamma)cR} \tag{1}$$

where T_a is the integration time, γ is the data overlap rate, ρ_a is the cross range resolution, f_c is the radar carrier frequency, υ is the velocity of the platform, c is the speed of light, and R is the radar range. The requirements on resolution and frame rate often lead the video SAR system to work at extremely high frequency (EHF) and even Terahertz band. Even if a recursive imaging algorithm can be used to generate more frames [29], the integration time for a frame of image should be short enough to monitor moving targets in time.

The position-shift and defocus of a moving target in SAR imagery are essentially caused by the phase mismatch between the target signature and the reference used in the imaging algorithm. Assume that a point moving target T locates at $(x_0, y_0, 0)$ at $t = 0$, the position shift and the defocus amount

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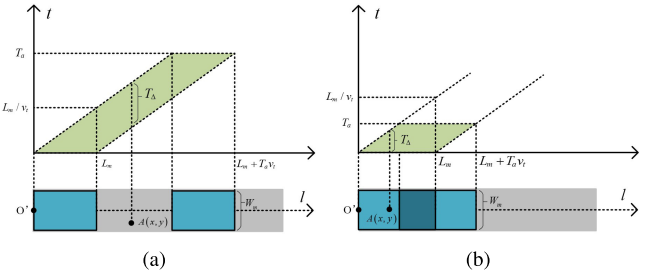

Fig. 2. Illustrations of occlusion time caused by a moving target. (a) $v_t > L_m/T_a$. (b) $v_t < L_m/T_a$.

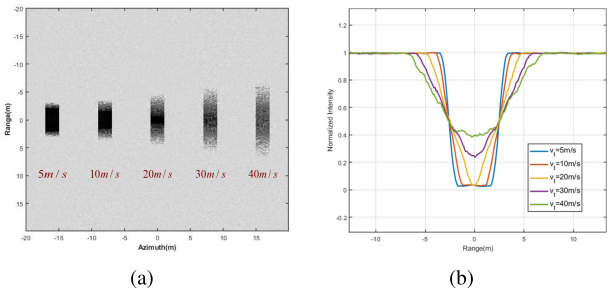

Fig. 3. Simulation of moving target shadows in a video SAR image. (a) Imaging results for targets with velocities being labeled below the corresponding shadows. (b) Normalized average intensities of the shadows along the moving direction.

of the smeared image [30], [31] can be expressed as

$$\Delta x = \frac{x_0 v_x + (y_0 + R_d) v_y}{v} \tag{2}$$

and

$$\delta_x = \frac{{v_x}^2 - 2v_x v + {v_y}^2 + (y_0 + R_d)a_y + x_0 a_x}{v} T_a + \frac{3(a_x v - a_x v_x - a_y v_y)}{4v} {T_a}^2 + \cdots \tag{3}$$

respectively, where the high-order terms of defocus amount can be neglected when T_a is small. v_x, v_y, a_x, a_y are the velocity and acceleration components of the target along the x-axis and y-axis, respectively. The position shift is sensitive to the radial velocity of the target, while the defocus amount is mainly determined by the target motion states and the integration time. Considering the short work range and integration time of video SAR, the image of a steady moving target mainly shows the position shift, and defocus appears when the radial acceleration is large.

The geometry of the shadow formation of a vehicle is also illustrated in Fig. 1, where the vehicle moves with a velocity of v_t from point ① and point ② during the integration period T_a. Ignore the shadow projected by the target height and assume that the length and width of the vehicle are L_m and W_m, respectively. The target could leave a shadow up to $L_m + T_a v_t$ long at its real position where the background was occluded by the target and the target image is shifted away. It can be grouped into two cases according to the target velocity along the moving track l, as shown in Fig. 2. The occlusion time T_Δ varies linearly along the moving track nearby the two ends of the occluded area. The occlusion time is a constant L_m/v_t in the middle area when $v_t > L_m/T_a$. A part of the middle area is completely occluded during the integration time and presents an ideal shadow when $v_t < L_m/T_a$, except for some inherent noise and possible interferences. It should be pointed that the height of the target could also project a shadow area on the ground, resulting in a positioning error for targets. This part is analyzed in Appendix A.

Fig. 3 shows the imaging result for a homogeneous scene including five simulated moving targets with the same size of 2.5 m × 5 m, but different velocities of 5, 10, 20, 30, and 40 m/s, respectively, for $f_c = 94$ GHz, $T_a = 0.2$ s, and $\rho_a = 0.1$ m. Fig. 3(a) shows the target shadows in the SAR image while all target images are shifted out of the scene due to the relative motion between the targets and the radar. Fig. 3(b) depicts the normalized average intensities of target shadows along the target moving direction. As seen, the slower the target, the more distinct the shadow will be. The shadow intensity roughly varies with the occlusion time linearly.

The shadow of a moving target varies with the parameters of the SAR system and target motion state. However, the shadow is not reliable for the target detection and tracking with acceptable false alarm and miss-detection rates due to the interference of target images and strong clutter.

III. Framework for Tracking Moving Target

We develop a joint framework that combines the shadow and echo energy information of moving targets for video SAR.

The returned signal from a moving target usually cannot be properly focused by SAR algorithms that are designed for the stationary scene. The echo energy of the target may be scattered in the image or even out of the imaging area. In this case, it is more likely to be detected in the RD domain with shorter integration time and full Doppler sampling bandwidth. In addition, the shadow-based moving target detection and tracking can be enhanced by associating it with the energy-based processing in the RD domain, which works perfectly in some cases, for example, when the target comes to a stop or the target shadow is smeared by interferences.

The proposed framework is shown in Fig. 4. First, the predetections for target shadow and energy are carried out in the SAR image and RD domain separately frame by frame. Both these predetections are designed with a high probability of detection and a high false alarm rate. Then, the M/N logic-based interframe data association and matching processing on the associated tracks between the two domains are performed to suppress the false alarms. Lastly, track update and management strategies are applied to deal with different cases of moving targets. This joint processing could significantly improve the robustness of detection and tracking for maneuvering targets.

A. Predetection of Target Shadow in SAR Imagery

In contrast to traditional target detection, shadow detection distinguishes low-level shadows from the high-level background. Since the speckle noise could affect the shadow detection severely, the detection of a single shadow pixel is not

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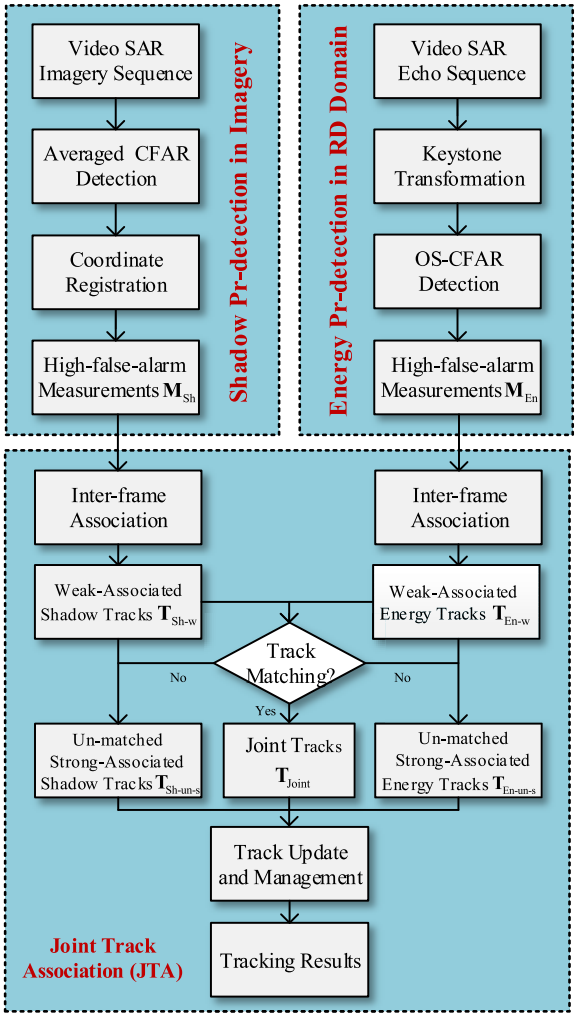

Fig. 4. Flowchart of the proposed framework for robust moving target tracking in video SAR.

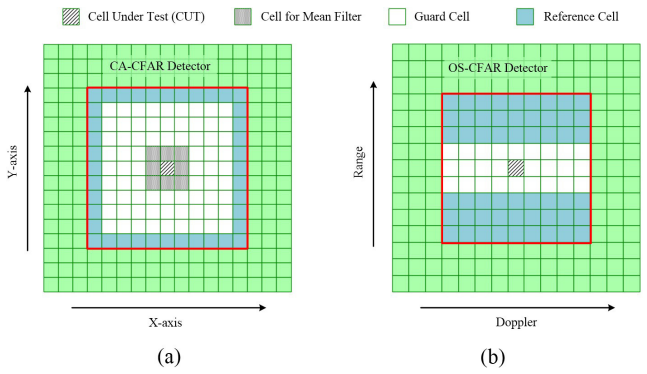

Fig. 5. CFAR predetectors based on target shadow and echo energy. (a) Shadow detector in SAR imagery. (b) Energy detector in the RD domain.

very meaningful. It has been shown that the median filtering can improve the shadow detection performance effectively at the expense of coarsening the resolution while preserving edges [32]. In our framework, a designed CFAR detector based on the mean filtering is used for a higher probability of detection, as shown in Fig. 5(a). To test if a cell (pixel) under test (CUT) belongs to a shadow, we take the mean of pixels in a window centered at the CUT as its test statistic. The size of the window is set to match that of the expected shadow. The cell-averaged CFAR (CA-CFAR) detector is then used for the shadow detection. The test statistic is compared with a threshold that is determined by the mean of reference cells around the CUT excluding some nearby guard cells. Specifically, the test statistic, that is, the output of the mean filter, for the (i_0, j_0)th CUT can be expressed as

$$C(i_0, j_0) = \frac{1}{N} \sum_{(i,j)\in S_N(i_0,j_0)} I(i, j) \tag{4}$$

where $S_N(i_0, j_0)$ denotes the index set of the pixels in the mean filter window centered at the (i_0, j_0)th CUT, N is the number of pixels in $S_N(i_0, j_0)$, and $I(i, j)$ denotes the intensity of the (i_0, j_0)th pixel. The intensity of each pixel in the SAR image obeys the negative exponential distribution with a mean equal to the backscatter intensity, which can be described by the equivalent backscatter coefficient (EBC) [33]. Under the assumption that the small background area around the shadow is homogeneous with the EBC being σ_B, the probability density function (PDF) of the test statistic x in background area can be calculated as

$$f_B(x) = \frac{N^{N-1} e^{-\frac{N}{\sigma_B} x} x^{N-1}}{{\sigma_B}^N \Gamma(N)} \tag{5}$$

where $\Gamma(\cdot)$ is the gamma function. For a given threshold A_T, the probability of false alarm (PFA) can be calculated as

$$P_{fa} = \int_0^{A_T} f_B(x)dx = e^{-NA_T/\sigma_B} \sum_{i=N}^{\infty} \frac{(NA_T/\sigma_B)^i}{i!}. \tag{6}$$

Reversely, the threshold A_T can be obtained from (6) for a given PFA. By taking $\sigma_B = -16.4$ dB, $L_m = 5$ m, and $T_a = 0.2$ s for an example, the receiver operating characteristic (ROC) curves for the single pixel in the shadow center area and the output after $N = 3 \times 3$ mean filtering with different target velocities are shown in Fig. 6. On the one hand, the target with lower velocity has better detection performance because of the longer occlusion time. On the other hand, the detector based on mean filtering improves the detection performance effectively by relieving the effect of speckle noise.

After the CA-CFAR detection, the morphological processing and connected area screening operation are performed on the binary image [34] according to the SAR resolution and shadow size. The coordinate registration is performed according to the imaging geometry to ensure the position alignment between frames. The registered centroid coordinates of the screened shadow area are obtained as the detection results for candidates of moving target in SAR imagery as

$$\mathbf{M}_{\mathrm{Sh}}(k) = \left\{ \boldsymbol{m}^i_{\mathrm{Sh}}(k) \right\}_{i=1}^{I_k} \tag{7}$$

where

$$\boldsymbol{m}^i_{\mathrm{Sh}}(k) = \left(x^i(k), y^i(k) \right)^T \tag{8}$$

and I_k is the number of predetections in the kth SAR image.

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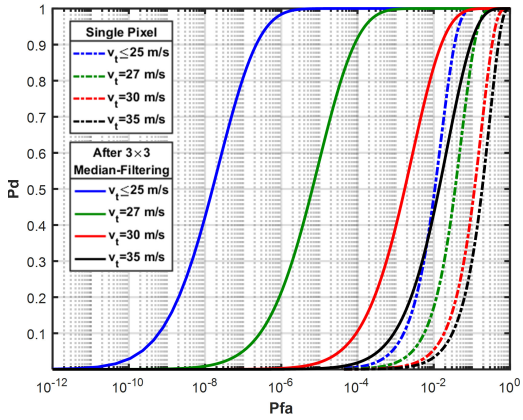

Fig. 6. ROC curves of the shadow detection with different target velocities. The target with lower velocity has better detection performance because of the longer occlusion time. The detector based on mean filtering can improve the detection performance effectively by relieving the effect of speckle noise.

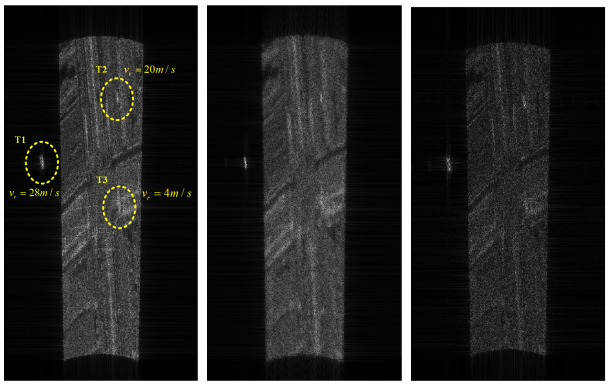

Fig. 7. RD spectra to be detected. (a) Original RD spectrum. (b) RD spectrum by keystone transformation. The range walk of the moving target is corrected effectively with ambiguity number equal to 2. (c) RD spectrum after clutter suppression. The SCRs of targets are improved from 42.67, 15.90, and 3.89 dB to 45.33, 20.88, and 4.46 dB, which is conducive to the subsequent detection.

B. Predetection of Target Energy in RD Domain

This section tries to find candidates for moving targets through intensity detection in the RD domain. Partial pulse echoes extracted from the center part of SAR data for each frame are used to obtain the RD spectrum by using the fast Fourier transform (FFT), where the coherent processing by FFT generally requires the range migration of observation scene not to exceed half of the range resolution in the integration time. The short integration time is also conducive to mitigate the defocus of moving targets according to the above analysis.

To achieve a high detection probability, the original RD spectrum with its processed images by the keystone transformation (KT) [35] and clutter suppression are fed into the detector simultaneously. The predetection results are the combination of detection results of these spectral images. The original RD spectrum preserves the strong reflection object to the greatest extent. The KT is applied to correct the possible range walk of moving targets for a higher positioning accuracy. The clutter suppression processing by noncoherent subimage subtraction in [36] and [37] is applied to relieve the influence of background clutter. By the noncoherent subtraction, the strong stationary clutters are more likely to be weaken, while the target energy can be retained due to its motion and fluctuation. The readers can refer to [36] and [37] for more details. An original RD spectrum with its processed images by KT and clutter suppression (from left to right) are shown in Fig. 7, where the radar carrier frequency $f_c = 94$ GHz, pulse repetition frequency $prf = 8$ kHz, 512 pulse echoes are used for FFT. Three targets with radial velocities of 28, 20, and 4 m/s are set in the scene and the corresponding signal to clutter ratios (SCRs) are 42.67, 15.90, and 3.89 dB, respectively. On the one hand, the range walk of the moving target is corrected effectively with an ambiguity number equal to 2 [see Fig. 7(b)]. On the other hand, the SCRs are improved to 45.33, 20.88, and 4.46 dB, respectively, by the clutter suppression processing [see Fig. 7(c)], which is conducive to the subsequent detection. Note that the target T3 is still difficult to be distinguished because of the slow velocity and strong clutter. This kind of targets often can be tracked by their shadows. The combined results of multiple processed spectral images will effectively improve the detection probability for target energy.

The order statistics (OS)-CFAR detector [38] has advantages in antiinterference, especially for the cases of multitarget and clutter edges. A modified OS-CFAR detector as shown in Fig. 5(b) is applied to the RD spectra with their processed images by KT and noncoherent clutter suppression simultaneously for a more reliable predetection. Considering that the energy of a moving target may smear, only the cells along the range dimension are taken as the reference cells to reduce the influence of the defocused energy of target. The reference cells are re-ordered as a sequence $\{I_{(1)}, I_{(2)}, \ldots, I_{(N_{\text{ref}})}\}$ according their intensities in ascending order. The nth-order statistics is selected as representative of the interference level and a threshold α_{os} is set as a multiple of this value

$$\xi_T = \alpha_{os} I_{(n)} \tag{9}$$

where the order n and the threshold factor α_{os} are chosen on an experimental basis.

The ROC curves of the modified OS-CFAR detector are given in Fig. 8. Assume that the background area is homogeneous and uses the Swerling 1 or Swerling 2 model for the moving target [39]. The signal to clutter and noise ratios (SCNRs) are set as 20 and 12 dB, which can represent the typical SCNRs in the exo- and endoclutters. By setting the number of reference cells N to 50 and 20, the choosing order n to 40, 25, 15, and 10, Fig. 8 shows that the detection performance improves with the increase of N and n. However, considering that the assumption of the homogeneous background area and the ability of antiinterference, N and n cannot be increased blindly. N is usually chosen by the resolution of the image for a proper background estimation and the order n can be approximately set to $0.75N$ for an acceptable performance [40]. The modified OS-CFAR detector has been shown to have a good performance in our experiments.

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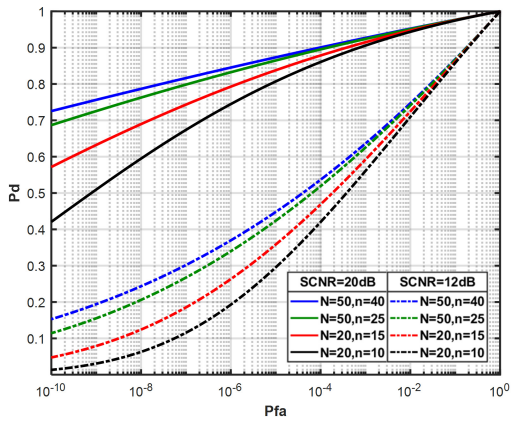

Fig. 8. ROC curves of the modified OS-CFAR detector. The detection performance improves with the increase of the number of reference cells N and the chosen order n under the assumption that the background area is homogeneous with no interferences.

In addition, the velocity ambiguity may occur for a high-speed target. Our strategy is to transform the original spectrum by using KT for all possible ambiguity numbers, send them to the CFAR detector, and then combine the detection results for a high detection probability. For a numeric example, the ambiguity number set $L \in [-2, 2]$ covers moving targets with velocities up to 30 m/s for an SAR system with $f_c =$ 94 GHz and $prf = 8$ kHz.

After the CFAR detection, the morphological processing and connected area screening operation are performed on the binary image [34] according to the spectral resolution and maximal detectable velocity. The centroid coordinates of the screened energy area are obtained as the detection results for candidates of moving target in RD domain as

$$\mathbf{M}_{\mathrm{En}}(k) = \left\{ \boldsymbol{m}_{\mathrm{En}}^{j}(k) \right\}_{i=1}^{J_k} \tag{10}$$

where

$$\boldsymbol{m}_{\mathrm{En}}^{j}(k) = \left(\mu f_a^j(k), r^j(k) \right)^T \tag{11}$$

where J_k is the number of predetections in the kth RD spectrum, $\mu = \lambda R_0/(2v)$ is the constant scale factor that transforms Doppler into azimuth scale for the consistency with range, and R_0 is the scene center slant range. The direct intensity detection effectively reduces the miss-detection rates for slowly moving targets, while the large number of false alarms generated by strong clutter will be suppressed in the subsequent processing.

C. Joint Track Association

Both the predetection methods based on the target shadow and echo energy are designed to have a high probability of detection, which leads to a high false alarm rate as well. Even so, the missed detections are inevitable because of the undeveloped shadows and defocused energy, which happens for maneuvering targets in video SAR from time to time. The joint track association (JTA) algorithm between SAR imagery and the corresponding RD spectra is proposed to suppress false alarms and reconstruct missed detections for a robust tracking.

Algorithm 1 JTA Algorithm

Input: $\mathbf{M}_{\mathrm{Sh}}(k) = \{\boldsymbol{m}_{\mathrm{SH}}^{i}(k)\}_{i=1}^{I_k}$
$\mathbf{M}_{\mathrm{En}}(k) = \{\boldsymbol{m}_{\mathrm{En}}^{j}(k)\}_{j=1}^{J_k} \quad k = 1, 2, \cdots K$

Initialization: $\mathbf{Z}_{\mathrm{Sh}}(1) = \mathbf{M}_{\mathrm{Sh}}(1)$, $\mathbf{Z}_{\mathrm{En}}(1) = \mathbf{M}_{\mathrm{En}}(1)$

Get $\mathbf{Z}_{\mathrm{Sh}}(2)$ and $\mathbf{Z}_{\mathrm{En}}(2)$ by NNS.

for k=2 to K **do**

State Estimation:

1. Get $\hat{\boldsymbol{v}}_{\mathrm{Sh}}^{m}(k)$ and $\hat{\boldsymbol{a}}_{\mathrm{Sh}}^{m}(k)$ $(k \geq 3)$ by kalman filtering.
2. Get $\hat{\boldsymbol{d}}_{\mathrm{En}}^{n}(k)$ by the first-order difference.

Prediction and Searching:

1. Get $\hat{\boldsymbol{z}}_{\mathrm{Sh}}^{m}(k+1)$ and $\hat{\boldsymbol{z}}_{\mathrm{En}}^{n}(k+1)$ by one-step prediction.
2. Get searching area $\mathbf{A}_{\mathrm{Sh}}^{m}(k+1)$ and $\mathbf{A}_{\mathrm{En}}^{n}(k+1)$.
3. Search measurements in $\mathbf{M}_{\mathrm{Sh}}(k+1)$ closest to $\hat{\boldsymbol{z}}_{\mathrm{Sh}}^{m}(k+1)$ in $\mathbf{A}_{\mathrm{Sh}}^{m}(k+1)$. If it fails, let $\boldsymbol{z}_{\mathrm{Sh}}^{m}(k+1) = \hat{\boldsymbol{z}}_{\mathrm{Sh}}^{m}(k+1)$.
4. Search measurements in $\mathbf{M}_{\mathrm{En}}(k+1)$ closest to $\hat{\boldsymbol{z}}_{\mathrm{En}}^{n}(k+1)$ in $\mathbf{A}_{\mathrm{En}}^{n}(k+1)$. If it fails, let $\boldsymbol{z}_{\mathrm{En}}^{n}(k+1) = \hat{\boldsymbol{z}}_{\mathrm{En}}^{n}(k+1)$.

Track Generation:

1. Change the logic value Γ=M/N in the previous N frames and get the strong- and weak-associated track set in imagery and RD domain ($\mathbf{T}_{\mathrm{Sh\text{-}s}}$, $\mathbf{T}_{\mathrm{En\text{-}s}}$, $\mathbf{T}_{\mathrm{Sh\text{-}w}}$ and $\mathbf{T}_{\mathrm{En\text{-}w}}$).

Track Association:

1. Project the $\mathbf{T}_{\mathrm{Sh\text{-}w}}$ to RD domain and get $\mathbf{T}_{\mathrm{pro}}$ according to the projecting relationship given by (27)(28)(29).
2. Match the $\mathbf{T}_{\mathrm{Sh\text{-}w}}$ with $\mathbf{T}_{\mathrm{pro}}$ by the range and Doppler constraints given by (31)(32).
3. Get the final output track set by uniting the joint track set and the un-matched strong-associated track sets: $\mathbf{T}_{\mathrm{Output}} = \mathbf{T}_{\mathrm{Joint}} \cup \mathbf{T}_{\mathrm{Sh\text{-}un\text{-}s}} \cup \mathbf{T}_{\mathrm{En\text{-}un\text{-}s}}$.

Track Update and Management:

1. Keep shadow tracking with the weak-association condition if the energy-based track is interrupted after the successful matching.
2. Reconstruct the missed measurements of shadow by searching pixels in candidate area to match the corresponding echo energy according to (35) if the shadow-based track is interrupted.
3. Identify and retain the tracks of go-and-stop targets by (36) for re-tracking.
4. Splice the intermittent tracks in imagery by (37) for consistent tracking.

end for

Output: The revised $\mathbf{T}_{\mathrm{Output}}$ after the update and management.

The measurements (i.e., predetection results) of target shadow and echo energy are associated between frames to screen potential moving targets by using the M/N logic-based method [41], separately. The logic-based method with a logic value $\Gamma :=$ M/N is used to screen out the potential moving targets. No less than M measurements in different frames are regarded as valid detections to form a candidate track if they are associated according to our designed state parameter constraints in N consecutive frames. The constraints on the state parameters will be discussed later. The unassociated measurements which are isolated or glint between frames and mainly produced by stationary objects can be suppressed effectively by the M/N logic-based method. However, it is hard to find a proper logic value Γ to make an assertive judgment on all measurements in practice. A low Γ involves too many false tracks, while a high Γ may cause failure in confirming the measurements of a moving target that appears intermittently, especially when the measurement error is large.

In our strategy, the weak-associated tracks for target shadow and echo energy with a low logic value $\Gamma_{\mathrm{w}} = \mathrm{M_w/N}$ are

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established first. The weak-association tries to find the possible tracks at a low missing rate. The isolated or glint objects between frames can be suppressed effectively. However, for the area with dense false alarms such as an area with many stationary objects being along the roadside, false tracks occur with the change of radar observation angle. Then, the JTA is conducted on the weakly associated tracks by matching the shadow tracks with the energy tracks to suppress the false tracks further and reconstruct the missed and interrupted tracks. Besides, considering that a mismatch may occur when the target shadow or energy disappears for a period of time, the strong association with a high logic value $\Gamma_s = M_s/N$ is conducted to track the target individually. The strong association generates tracks with high confidence and low false alarm rate for the supplement to the final tracking. Lastly, the track update and management are performed for the track judgment and reconstruction in different cases of moving targets. Starting from the matched weak association, then fusing with the strong association, followed by track update and management, this joint track-association processing provides a more robust moving target tracking method for video SAR comprehensively. The JTA algorithm is presented by Algorithm 1 and detailed as follows.

Consider the predetection results based on the target shadow and echo energy as expressed in (7) and (10). Let $\mathbf{Z}_{\mathrm{Sh}}(1) = \mathbf{M}_{\mathrm{Sh}}(1)$, $\mathbf{Z}_{\mathrm{En}}(1) = \mathbf{M}_{\mathrm{En}}(1)$, and start the tracks by using the nearest the neighbor search (NNS) method [42]. In the kth frame, define the tracks of target shadow and echo energy in the imagery and the corresponding RD domain as

$$\mathbf{Z}_{\mathrm{Sh}}(k) = \left\{z_{\mathrm{Sh}}^m(k)\right\}_{m=1}^{M_k} \tag{12}$$

and

$$\mathbf{Z}_{\mathrm{En}}(k) = \left\{z_{\mathrm{En}}^n(k)\right\}_{n=1}^{N_k} \tag{13}$$

respectively, where M_k and N_k denote the numbers of target tracks included in the kth frame of the imagery and the corresponding RD domain, respectively.

For the shadow-based target tracking in imagery, the Kalman filter is used to estimate the state vectors of tracks. The motion model used for target shadow is the constant acceleration (CA) model [43], which has been verified to have a good performance in the track update for shadows by using the Kalman filter. The estimated state vector for the mth track in the kth frame is defined as

$$\boldsymbol{x}^m(k) := [x^m(k), v_x^m(k), a_x^m(k), y^m(k), v_y^m(k), a_y^m(k)]^T. \tag{14}$$

The state transition equation is given by

$$\boldsymbol{x}^m(k+1) = \Phi \boldsymbol{x}^m(k) + \boldsymbol{W}(k) \tag{15}$$

where Φ and $\boldsymbol{W}(k)$ represent the state transition matrix and zero-mean white Gaussian noise, respectively. The instant velocity and acceleration of the mth target in the kth frame are estimated by

$$\hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k) = \left[v_x^m(k), v_y^m(k)\right]^T \tag{16}$$

and

$$\hat{\boldsymbol{a}}_{\mathrm{Sh}}^m(k) = \left[a_x^m(k), a_y^m(k)\right]^T \tag{17}$$

respectively. The predicted position of the target and the searching area in next frame are given by

$$\hat{z}_{\mathrm{Sh}}^m(k+1) = z_{\mathrm{Sh}}^m(k) + \hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k)/fps + \frac{1}{2}\hat{\boldsymbol{a}}_{\mathrm{Sh}}^m(k)/fps^2 \tag{18}$$

and

$$\mathbf{A}_{\mathrm{Sh}}^m(k+1) = \left\{ (x,y)^T \left| \begin{array}{l} \left\|\boldsymbol{v}_{\mathrm{Sh}}(k+1) - \hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k)\right\|_2 < \alpha_v \left\|\hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k)\right\|_2 \\ \left\|\boldsymbol{a}_{\mathrm{Sh}}(k+1) - \hat{\boldsymbol{a}}_{\mathrm{Sh}}^m(k)\right\|_2 < \alpha_a \left\|\hat{\boldsymbol{a}}_{\mathrm{Sh}}^m(k)\right\|_2 \\ \arccos\left(\dfrac{\boldsymbol{v}_{\mathrm{Sh}}(k+1)^T \cdot \hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k)}{\|\boldsymbol{v}_{\mathrm{Sh}}(k+1)\|_2 \left\|\hat{\boldsymbol{v}}_{\mathrm{Sh}}^m(k)\right\|_2}\right) < \Delta\theta_{\mathrm{Sh}} \end{array} \right. \right\} \tag{19}$$

where $\boldsymbol{v}_{\mathrm{Sh}}(k+1)$ and $\boldsymbol{a}_{\mathrm{Sh}}(k+1)$ are the velocity and acceleration calculated from $(x,y)^T$, $z_{\mathrm{Sh}}^m(k)$, and $z_{\mathrm{Sh}}^m(k-1)$, α_v and α_a are the variation factors of velocity and acceleration, respectively, and $\Delta\theta_{\mathrm{Sh}}$ is the maximum searching angle. The state parameter set $\mathbf{P}_{\mathrm{Sh}} = \{\alpha_v, \alpha_a, \Delta\theta_{\mathrm{Sh}}\}$ determines the degrees of constraints and is chosen on an experimental basis. The acceleration constraint in (19) is neglected if the number of associated measurements is less than 3. Then, the update for the next frame is determined by a measurement in $\mathbf{A}_{\mathrm{Sh}}^m(k+1)$ nearest to $\hat{z}_{\mathrm{Sh}}^m(k+1)$. If no measurements are found, the update is given by $\hat{z}_{\mathrm{Sh}}^m(k+1)$, that is,

$$z_{\mathrm{Sh}}^m(k+1) = \begin{cases} \underset{\boldsymbol{m}_{\mathrm{Sh}}(k+1)\in\mathbf{A}_{\mathrm{Sh},k+1}^m}{\operatorname{argmin}} \left(\left\|\boldsymbol{m}_{\mathrm{Sh}}(k+1) - \hat{z}_{\mathrm{Sh}}^m(k+1)\right\|_2\right), & \text{if exist} \\ \hat{z}_{\mathrm{Sh}}^m(k+1), & \text{otherwise.} \end{cases} \tag{20}$$

Similarly, for target energy in the RD domain, we estimate the state vector by the first-order difference. The difference and one-step prediction are given by

$$\hat{\boldsymbol{d}}_{\mathrm{En}}^n(k) = \boldsymbol{m}_{\mathrm{En}}^n(k) - \boldsymbol{m}_{\mathrm{En}}^n(k-1) \tag{21}$$

and

$$\hat{z}_{\mathrm{En}}^n(k+1) = z_{\mathrm{En}}^n(k) + \hat{\boldsymbol{d}}_{\mathrm{En}}^n(k) \tag{22}$$

respectively. The searching area and update for the next frame are

$$\mathbf{A}_{\mathrm{En}}^n(k+1) = \left\{ (f_a, r)^T \left| \begin{array}{l} \left\|\boldsymbol{d}_{\mathrm{En}}(k+1) - \hat{\boldsymbol{d}}_{\mathrm{En}}^n(k)\right\|_1 < \alpha_d \left\|\hat{\boldsymbol{d}}_{\mathrm{En}}^n(k)\right\|_1 \\ \arccos\left(\dfrac{\boldsymbol{d}_{\mathrm{En}}(k+1)^T \cdot \hat{\boldsymbol{d}}_{\mathrm{En}}^n(k)}{\|\boldsymbol{d}_{\mathrm{En}}(k+1)\|_2 \left\|\hat{\boldsymbol{d}}_{\mathrm{En}}^n(k)\right\|_2}\right) < \Delta\theta_{\mathrm{En}} \end{array} \right. \right\} \tag{23}$$

and

$$z_{\mathrm{En}}^n(k+1) = \begin{cases} \underset{\boldsymbol{m}_{\mathrm{En}}(k+1)\in\mathbf{A}_{\mathrm{En}}^n(k+1)}{\operatorname{argmin}} \left(\left\|\boldsymbol{m}_{\mathrm{En}}(k+1) - \hat{z}_{\mathrm{En}}^n(k+1)\right\|_1\right), & \text{if exist} \\ \hat{z}_{\mathrm{En}}^n(k+1), & \text{otherwise} \end{cases} \tag{24}$$

respectively, where α_d is the variation factor of first-order difference, and $\Delta\theta_{\mathrm{En}}$ is the maximum searching angle in

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the RD domain. The state parameter set $\mathbf{P}_{\mathrm{En}} = \{\alpha_d, \Delta\theta_{\mathrm{En}}\}$ determines the degrees of constraints in the RD domain and is also chosen on an experimental basis.

After the prediction and searching for target shadow and echo energy, the candidate tracks are classified according to the logic value Γ frame by frame. For the given weak-association logic value $\Gamma_{\mathrm{w}} = \mathrm{M_w/N}$ and the strong-association logic value $\Gamma_{\mathrm{s}} = \mathrm{M_s/N}$, we count the number of measurements from effective searching in the candidate tracks in the previous N frames and obtain the weak and strong associated track set of the shadow and energy ($\mathbf{T}_{\text{Sh-w}}$, $\mathbf{T}_{\text{En-w}}$, $\mathbf{T}_{\text{Sh-s}}$, and $\mathbf{T}_{\text{En-s}}$), respectively. Obversely, we have the relationship that $\mathbf{T}_{\text{Sh-s}} \subseteq \mathbf{T}_{\text{Sh-w}}$ and $\mathbf{T}_{\text{En-s}} \subseteq \mathbf{T}_{\text{En-w}}$. Γ_{w} and Γ_{s} are determined by the background clutter and predetection condition in a strict sense. Considering that the complexity of the scene and the limited N, Γ_{w} and Γ_{s} are suggested to be chosen by experimental means. Usually, Γ_{w} less than 0.5 and Γ_{s} greater than 0.75 can be used as a reference for the ideal tracking performance.

The joint processing is then applied by matching the weak-associated tracks of target shadows with those of echo energy to suppress the false tracks and reconstruct the missed detections. The pth weak-associated track of shadow and the qth weak-associated track of energy in the kth frame can be expressed as

$$\mathbf{T}^{p}_{\mathrm{Sh-w},k} = \left\{ z^{p}_{\mathrm{Sh}}(l) = (x^{p}(l), y^{p}(l)) \right\}^{k}_{l=k^{p}_{\text{Sh-start}}} \tag{25}$$

and

$$\mathbf{T}^{q}_{\mathrm{En-w},k} = \left\{ z^{q}_{\mathrm{En}}(l) = (\mu f^{q}_{a}(l), r^{q}(l)) \right\}^{k}_{l=k^{q}_{\text{En-start}}} \tag{26}$$

respectively, where $k^{p}_{\text{Sh-start}}$ and $k^{q}_{\text{En-start}}$ represent the starting frame for the pth track of shadow and the qth track of energy, respectively. The weak-associated tracks of shadow are then projected to the RD domain for distance matching.

The projected range is calculated by the distance between radar and track points, that is,

$$r^{p}_{\mathrm{pro}}(l) = \left\| z^{p}_{\mathrm{Sh}}(l) - \boldsymbol{p}(l) \right\|_2 \tag{27}$$

for $l = k-N+1, \ldots, k$, where $\boldsymbol{p}(l)$ is the radar position vector in the lth frame defined in Fig. 1. The projected Doppler is calculated by the relative velocity v_r between radar and track points according to $f_d = (2/\lambda)v_r$. According to the geometry in Fig. 1, the relative velocity v_r can be expressed as the projection of the velocity vector sum of the radar and the target on their straight line. Therefore, the projected Doppler in the RD domain can be expressed as

$$\begin{aligned} \tilde{f}^{p}_{\mathrm{a,pro}}(l) &= \frac{2}{\lambda}\boldsymbol{p}^{0}(l) \cdot \left[\boldsymbol{v}(l) - \hat{\boldsymbol{v}}^{p}_{\mathrm{Sh}}(l) \right] \\ &= \frac{2}{\lambda} \frac{\left[z^{p}_{\mathrm{Sh}}(l) - \boldsymbol{p}(l) \right] \cdot \left[\boldsymbol{v}(l) - \hat{\boldsymbol{v}}^{p}_{\mathrm{Sh}}(l) \right]}{\left\| z^{p}_{\mathrm{Sh}}(l) - \boldsymbol{p}(l) \right\|_2} \end{aligned} \tag{28}$$

where $\boldsymbol{p}^{0}(l)$ denotes the unit vector of the radar pointing to shadow in the lth frame, and $\boldsymbol{v}(l)$ and $\hat{\boldsymbol{v}}^{p}_{\mathrm{Sh}}(l)$ are the velocity vector of the radar and the pth track in the lth frame, respectively. Considering the Doppler ambiguity, the observed value of Doppler in the RD domain is

$$f^{p}_{\mathrm{a,pro}}(l) = \mathrm{mod}\left[\tilde{f}^{p}_{\mathrm{a,pro}}(l) + \frac{prf}{2}, prf \right] - \frac{prf}{2} \tag{29}$$

where mod(a, b) returns the remainder after division of a by b, and prf denotes the radar pulse repetition frequency. The projected track can be expressed as

$$\mathbf{T}^{p}_{\mathrm{pro},k} = \left\{ z^{p}_{\mathrm{pro}}(l) = (\mu f^{p}_{\mathrm{a,pro}}(l), r^{p}_{\mathrm{pro}}(l)) \right\}^{k}_{l=k^{p}_{\text{Sh-start}}}. \tag{30}$$

For any given p and q, $\mathbf{T}^{p}_{\mathrm{pro,k}}$ and $\mathbf{T}^{q}_{\mathrm{En-w,k}}$ are considered to be matched when

$$\sum_{l=k-N+1}^{k} \left\| r^{p}_{\mathrm{pro}}(l) - r^{q}(l) - \Delta r \right\|_1 < N\xi_r \tag{31}$$

$$\sum_{l=k-N+1}^{k} \left\| f^{p}_{\mathrm{a,pro}}(l) - f^{q}_{a}(l)) \right\|_1 < N\xi_{f_a} \tag{32}$$

where ξ_r and ξ_{f_a} are the matching thresholds for the range and Doppler, respectively, and Δr is the range inconsistency between the shadow projection and the energy. The formulas (31) and (32) ensure that the two tracks obtained from the target shadow and energy are consistent in the range and Doppler cells. The determination criteria for ξ_r and ξ_{f_a} can be found in Appendix B. It should be noted that the constraints are quite strict because they are carried out on multiple successive detections in the track simultaneously. Therefore, the false alarms can be eliminated effectively in this step. In addition, some missed tracking may be caused by the mismatching due to the positioning error of shadow and energy. This part of missed tracking can be complemented by the interframe strong association and the following track update and management. The joint track set of moving targets after matching can be expressed as

$$\mathbf{T}_{\mathrm{Joint}} = \left\{ \mathbf{T}^{p}_{\mathrm{Sh-w},k} \cup \mathbf{T}^{q}_{\mathrm{En-w},k} | p, q \ s.t. \ (31)(32) \right\}^{K}_{k=1}. \tag{33}$$

The matching processing for the weak association makes full use of the shadow and energy information for the confirmation of tracks and reduces the false alarms occurred after the predetection effectively. Moreover, the unmatched strong-associated track sets $\mathbf{T}_{\text{Sh-un-s}}$ and $\mathbf{T}_{\text{En-un-s}}$ also provide the target tracks in the imagery and RD domain with a high confidence level and low false rate separately. The output on the target tracking is a union of the joint track set and the unmatched strong-associated track sets, that is,

$$\mathbf{T}_{\mathrm{Output}} = \mathbf{T}_{\mathrm{Joint}} \cup \mathbf{T}_{\text{Sh-un-s}} \cup \mathbf{T}_{\text{En-un-s}}. \tag{34}$$

The JTA processing tracks moving targets at a low false alarm rate and missing rate simultaneously by combining detection results of the target shadow and echo energy. The missed detections are reconstructed by the one-step-prediction between frames by using (18) and (22). However, the one-step prediction is only suitable for the case that the missed measurements are sparse in adjacent frames and the measurement error is small. When a missed detection occurs in several consecutive frames, the prediction accuracy will be greatly reduced and the matching results will be affected, especially for maneuvering targets. Some missed tracks caused by the mismatching should also be reconstructed.

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D. Track Update and Management

The missed detections in the predetection are mainly due to the undeveloped or smeared shadows in the imagery and the defocused or submerged energy in the RD domain because of the complex environment and relative motion between the target and the radar. In our framework, the missed detections are reconstructed by interactive processing between the target shadow and echo energy with the update of tracks. Specifically, the track update and management methods are proposed for different situations on the existing target tracks and measurements. Several typical situations and the corresponding procedures for the track update and management are discussed as follows.

1) The shadow-based track for a moving target keeps going continuously while the energy-based track is interrupted after the successful matching processing.
   This situation often happens when the echo energy of the target is largely defocused or the energy is submerged in the strong clutter. In this case, the shadow-based track will continue by letting the weak-association condition be an effective tracking in the future frames rather than a strong-association condition, because the shadow-based track has been already confirmed by the matching processing as a real moving target.
2) The energy-based track for a moving target continues, while the shadow-based track is interrupted after the successful matching processing. This situation can be divided into two cases as follows.
   (a) *Continuously moving targets*: The shadow-based track could be interrupted when the target moves too fast to form a distinct shadow in the SAR imagery, or the target shadow is smeared by interferences such as those from the sidelobe leakage of strong clutter, a partial image of its own, or images of other moving targets. For a pair of matched tracks in the two domains, the missed measurement of the shadow-based track will be reconstructed by the corresponding measurement from the matched energy-based track in the same frame. Specifically, the missed measurements are reconstructed by searching the pixels in the candidate area predicted by (19) to replace the interframe prediction given by (20), that is,

$$z_{\mathrm{Sh}}^{p}(k+1)=\underset{\boldsymbol{m}_{\mathrm{Sh}}(k+1)\in\mathbf{A}_{\mathrm{Sh},k+1}^{p}}{\operatorname{argmin}}\left\|\boldsymbol{m}_{\mathrm{pro}}(k+1)-z_{\mathrm{En}}^{q}(k+1)\right\|_{1} \tag{35}$$

   where $\boldsymbol{m}_{\mathrm{pro}}(k+1)$ represents the projection of $\boldsymbol{m}_{\mathrm{Sh}}(k+1)$ from the imagery to the RD domain as expressed by (27)–(29). The recovered motion states of the targets by the joint processing have been found to have a good accuracy for tracking in experiments.
   (b) *Go-and-stop targets*: The echo energy will gradually be focused and covers on the target shadow when a moving target slows down and stops. The track of a stopped target should be retained so that the target can be re-tracked when it moves again. The go-and-stop target is identified if the estimated velocity is small enough, and its corresponding echo energy keeps detected in a stable area. Specifically, given velocity and distance thresholds V_T and D_T, the deciding conditions for a go-and-stop target can be expressed as

$$\begin{cases}\|\hat{\boldsymbol{v}}_{\mathrm{Sh}}^{q}(k_{\mathrm{end}})\|_2\le V_T\\ \forall k=k_{\mathrm{end}+1},\ldots,k_{\mathrm{end}+\mathrm{n}_0},\exists\boldsymbol{m}_{\mathrm{En}}(k)\\ \|\boldsymbol{m}_{\mathrm{En}}(k)-z_{\mathrm{En}}^{q}(k_{\mathrm{end}})\|_1\le D_T\end{cases} \tag{36}$$

   where $\hat{\boldsymbol{v}}_{\mathrm{Sh}}^{q}(k_{\mathrm{end}})$ denotes the velocity estimation in the last frame for the qth matched track, and $z_{\mathrm{En}}^{q}(k_{\mathrm{end}})$ denotes the last measurement for the qth matched track in the RD domain. The track will be retained in the following n_0 frames after the k_{end}th frame for re-tracking.
3) Both the shadow-based and energy-based tracks are lost for a period of time: This situation often happens for a maneuvering target when its echo energy is severely defocused and the shadow is undeveloped or smeared by interferences. The lack of shadow and energy information makes the tracks intermittent. For this case, the track splicing processing is performed for consistent tracking of the moving target in the image domain by matching the interrupted tracks with newly started tracks. Define a mismatched time T_{mis} (2 s is reasonable for a ground maneuvering target). Two tracks are considered to be from the same target if the following constraints for the position and direction angle of the target are met

$$\begin{cases}\left\|z_{\mathrm{Sh}}^{q}(k_{\mathrm{start}})-z_{\mathrm{Sh}}^{p}(k_{\mathrm{end}})\right\|_1\le v_{\max}T_{\mathrm{mis}}\\ \left\|\theta_{\mathrm{Sh}}^{q}(k_{\mathrm{start}})-\theta_{\mathrm{Sh}}^{p}(k_{\mathrm{end}})\right\|_1\le\omega_{\max}T_{\mathrm{mis}}\end{cases} \tag{37}$$

   where $\theta_{\mathrm{Sh}}^{q}(k_{\mathrm{start}})$ and $\theta_{\mathrm{Sh}}^{p}(k_{\mathrm{end}})$ denote the estimated direction angles in the newly started frame of qth track and the ended frame of the pth track, $v_{\max}$ and $\omega_{\max}$ denote the maximal velocity and angular velocity, respectively.

In summary, the track update and management strategies discussed above make a consistently track moving targets in different situations. In the proposed framework, false alarms are suppressed by combining the target shadow and echo energy information between frames, and missed detections can be reconstructed through joint processing between the shadow-based and energy-based tracks, resulting in a robust process for tracking moving targets in a video SAR.

E. Feasibility Analysis

The positioning error of both the target shadow and energy affect the data association. The feasibility of the association processing between moving target shadow and energy is analyzed in this section.

The motion state estimation for target shadow is based on a unified ground reference and the coordinate registration is conducted before the interframe association. Algorithms such as back projection algorithm (BPA) and polar formation algorithm (PFA) can be used for imaging. Each time a new frame of image is generated, the previous predetections are

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aligned to the new image according to the imaging geometry for the unified target positioning and motion state estimation. The image registration method such as scale-invariant feature transform (SIFT) algorithm [44] can also be used to achieve the unified positioning. Therefore, the shadow positioning accuracy is mainly affected by the height projection error and measurement error of image processing as discussed in Appendix A. For the energy positioning, the range positioning is quite accurate because the echo is calibrated strictly by the system receiving delay, except for some measurement errors of image processing. The Doppler accuracy is relatively low because it mainly depends on the pulse number for FFT and the defocusing degree of the target. Fortunately, the short work range and integration time of video SAR could mitigate the defocus for a moving target as analyzed by (3).

For an effective association, proper matching thresholds ξ_r and ξ_{f_a} derived in Appendix B are used to tolerate the positioning error for range and Doppler as given by formulas (31) and (32). It should be noted that the constraints for matching are quite strict (especially for the range) because they are carried out on multiple successive detections in the track for range and Doppler simultaneously. The experiments based on simulation and real data in Section IV have shown the effective association between the target shadow and energy.

In extreme cases, some maneuvering targets maybe seriously defocused as a bright line and completely submerged in ground clutter, resulting in an inaccurate/failed positioning for Doppler center. These inaccurate/missed Doppler centers cannot be associated and the target will be tracked by its shadow individually. Even so, the previous successful association can also improve the shadow tracking confidence, as given by Case 1) in Section III-D. When the target energy can be detected again, the joint processing continues. For a more accurate positioning, one can process the defocused energy by refocusing methods like in [35] and [45]. In fact, it is not necessary to keep the accurate detection for energy or shadow all the time during the tracking.

F. Computational Complexity

In this section, the computational complexity of the proposed framework is analyzed in detail. In predetection, the computational complexity for shadow detection per frame is of the order $O\left(N_x^{\text{im}} N_y^{\text{im}} \left(N_{\text{ref}}^{\text{im}} + N_m^{\text{im}}\right) + N_{pix,k}^{\text{im}} + I_k\right)$, where N_x^{im} and N_y^{im} denote the size of the image to be detected, $N_{\text{ref}}^{\text{im}}$ and N_m^{im} denote the number of reference cells and cells for mean filter of the CA-CFAR detector, $N_{pix,k}^{\text{im}}$ and I_k denote the number of detected pixels and connected areas in the kth frame. The three terms in the outer brackets represent the complexity of CFAR detection, morphological processing, and coordinate registration by rotation and/or translation. The computational complexity for energy detection per frame is of the order $O\left(N_x^{\text{rd}} N_y^{\text{rd}} N_{\text{ref}}^{\text{rd}\,2} + N_{pix,k}^{\text{rd}} + N_r N_D + N_y^{\text{rd}} N_s\right)$, where N_x^{rd} and N_y^{rd} denote the size of the RD spectrum to be detected, $N_{\text{ref}}^{\text{rd}}$ and $N_{pix,k}^{\text{rd}}$ denote the number of reference cells and detected pixels in the kth spectrum, N_r and N_D denote the sampling number of the echo in range and Doppler direction, N_s denotes the azimuthal size of the subimage for noncoherent subtraction. The four terms represent the complexity of CFAR detection, morphological processing, KT processing, and clutter suppression.

In JTA algorithm, the predetections are first associated between frames. The computational complexity of kth interframe association for shadow is of the order $O(M_{k-1} + I_k M_{k-1})$, where M_{k-1} denotes the number of established shadow tracks in the $(k-1)$th frame. The first term represents the complexity of the motion state estimation and the second represents that of the searching and update. Similarly, the computational complexity of kth interframe association for energy is of the order $O(N_{k-1} + J_k N_{k-1})$, where N_{k-1} denotes the number of established energy tracks in the $(k-1)$th frame. For projection and matching processing, the computational complexities of the kth frame are of the order $O(M_k N)$ and $O(M_k N_k N)$, respectively, where N denotes the number of frames for association.

The overall computational complexity of the proposed framework can be expressed as $O(N_x^{im} N_y^{im} (N_{\text{ref}}^{im} + N_m^{im}) + N_x^{rd} N_y^{rd} N_{\text{ref}}^{rd\,2} + N_r N_D + I_k M_{k-1} + J_k N_{k-1} + M_k N_k N)$. The calculation is mainly at the predetection stage and is related to the size of the detected image and the CFAR detector. Considering that the video SAR often works with a small observation scene, the computation will not too large. An intuitive example will be presented by the processing of real data.

TABLE I
PARAMETERS FOR VIDEO SAR SIMULATION

Parameter Name	symbol	value	unit
Center Frequency	f_c	94	GHz
Bandwidth	B_w	5	GHz
Pulse Duration	T_p	50	μs
Sampling Frequency	F_s	120	MHz
Pulse Repetition Frequency	prf	8000	Hz
Depression Angle	θ	53	$^\circ$
Flight Height	H	600	m
Flight Velocity	v	35	m/s
Single-frame Pulse Numbers	N_a	1946	-
Data Overlap Factor	γ	58.9	%
SAR Video Frame Rate	fps	10	Hz
Doppler Pulse Numbers	N_D	512	-
Scene Size	S_s	150×150	m×m
Clutter-to-Noise Ratio	CNR	13	dB
Signal-to-Clutter Ratio	SCR	8	dB

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the effectiveness of the proposed framework are evaluated by the simulated data and the real date released by the Sandia National Laboratory. The processing results with quantitative analysis are discussed in detail.

A. Results of Simulated Data

A set of simulated video SAR date is used to describe the process of the framework and various cases analyzed above. The video SAR for the simulations works in a circular imaging mode, with system and scenario parameters listed in Table I. The raw video SAR data are generated based on a real SAR reflectivity map that contains a crossroad. ISAR images of three typical vehicles (subcompact, sedan and truck) [46] with

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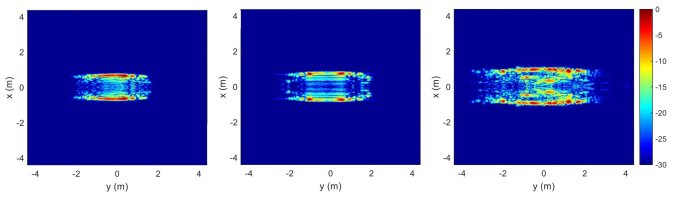

(a) (b) (c)

Fig. 9. ISAR images of three typical vehicles [46] used for the target models in the simulation. The orientations of the targets are adjusted according to their moving directions. (a) Subcompact with a size of 4 m×1.8 m. (b) Sedan with a size of 4.6 m×2 m. (c) Truck with a size of 5.2 m×2.2 m.

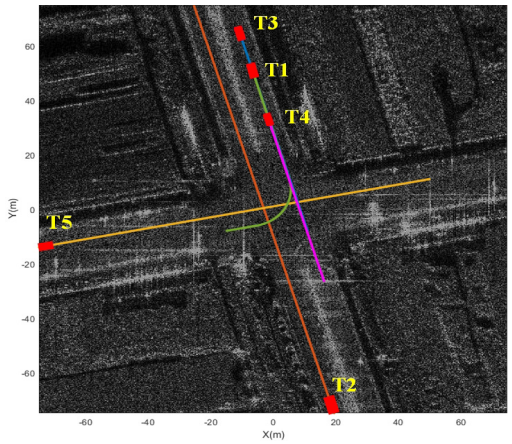

Fig. 10. Moving tracks of the simulated targets. Targets have typical motion parameters that meet the actual situations.

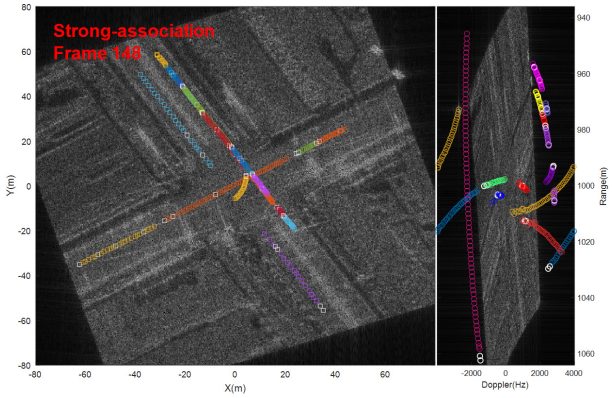

Fig. 11. Intermittent tracks formed by the strong association. The tracks of target shadow and energy are with a high confidence level and low false alarm rate. The processing results with high confidence can be used as a supplement to the tracking.

different sizes (4 m × 1.8 m, 4.6 m × 2 m, and 5.2 m × 2.2 m) are used as the target models for the simulations as shown in Fig. 9. The orientations of the targets are adjusted according to their moving directions. Fig. 10 shows the moving tracks of five simulated vehicle targets with motion parameters listed in Table II. The target echoes are injected into the background echo. The shadow is formed as a black area with the same shape and size as the area occluded by moving target, which is a good approximation for video SAR imaging with a large depression angle.

TABLE II
MOTION PARAMETERS OF SIMULATED VEHICLES

Acceleration along X and Y: a_x and a_y (m/s^2) Initial velocity along X and Y: v_x and v_y (m/s)					
Initial position (x_0, y_0)	Slow-time	v_x	v_y	a_x	a_y
T1 (-15,-8)	0 s ∼ 1.95 s	5	1	0	0
	1.95 s ∼ 6.35 s	Circular motion at 5.1 m/s.			
	6.35 s ∼ 12 s	-1.44	4.9	0	0
T2 (-39,120)	0 s ∼ 15 s	6	-20	0	0
T3 (11,-8)	0 s ∼ 15 s	-1.5	5	0	0
T4 (16.2,-26.4)	0 s ∼ 4.8 s	-1.8	6	0.375	-1.25
	4.8 s ∼ 7.5 s	0	0	0	0
	7.5 s ∼ 15 s	0	0	-0.55	1.83
T5 (50,11)	0 s ∼ 1.2 s	-6	-1.2	0.5	0.1
	1.2 s ∼ 13.2 s	-5.4	-1.08	-0.8	-0.16
	13.2 s ∼ 15 s	-15	-3	0	0

Target type: T1,T3: sedan; T2: truck; T4,T5: subcompact.

A total of 148 sequential frames of images and corresponding RD spectra are obtained from the simulated video SAR data by using the polar format algorithm and FFT at a frame rate of 10 Hz. In the SAR imagery, the target shadows are observable when the defocused images are shifted away from their real positions due to the relative motion between targets and radar. A moving target can be partially focused when its velocity is slow enough, and a higher velocity makes the shadow indistinctive because of the shorter occlusion time of the background. In the RD spectra, the scene is focused roughly with a large Doppler band, where the shifted and smeared echo energy of the moving targets can be found.

Fig. 11 shows the tracks obtained through the strong association with the logic value $\Gamma_s = 8/10$ in the image domain and the RD domain. As seen, the tracks of shadow-based and energy-based are with a high confidence level and low false alarm rate. However, the harsh formation conditions for the strong association make the tracks intermittent. The detection results with high confidence can be used as a supplement to the tracking.

Taking frames 24, 49, 94, and 128 as examples, we demonstrate the target tracking process for the moving targets using the proposed framework, as shown in Figs. 12 and 13. In Fig. 12(a)–(d) shows the SAR images and RD spectra with the predetection results for frames 24, 49, 94, and 128, respectively. The correct detections are marked in yellow and false alarms in cyan, where the five targets are numbered with yellow text. The predetection has a high detection probability, but at the cost of a large number of false alarms. Fig. 12(e)–(h) shows the data association results using the M/N logic-based method, where the cumulative weak-associated tracks until the current frame are marked in different colors with a weak logic value $\Gamma_w = 3/10$. The white marks represent the reconstructed measurements after taking the interframe prediction or the joint processing between shadow-based tracks and energy-based tracks. The weak-association processing generates candidate tracks at a high probability of detection. In Fig. 13(a)–(d), the tracks in imagery are projected to the

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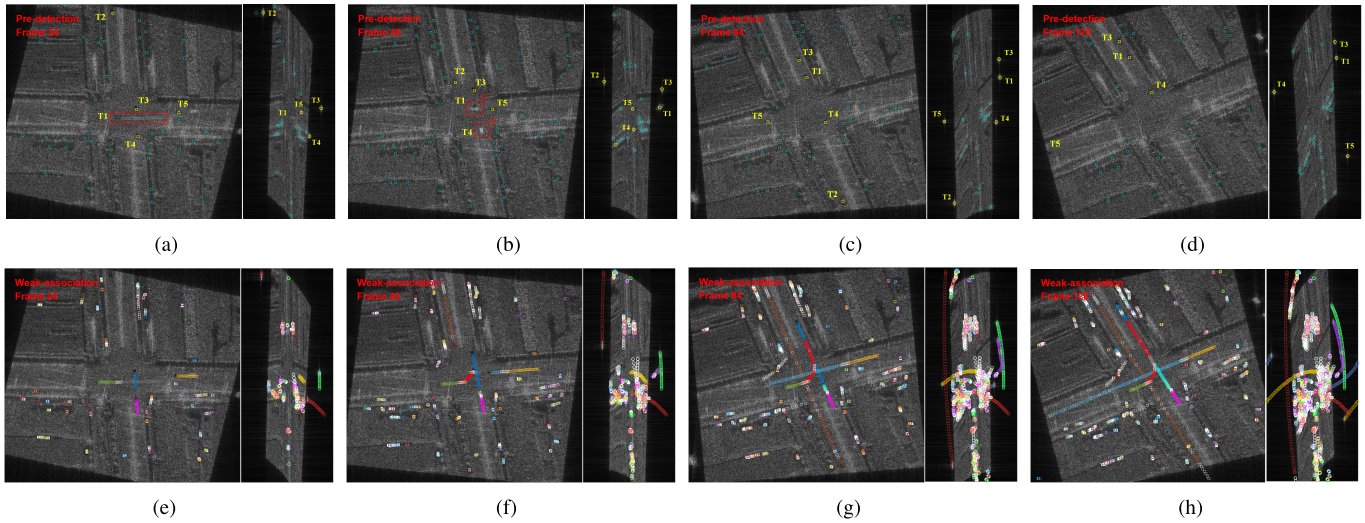

Fig. 12. Predetection and weak-association results in frames 24, 49, 94, and 128 of the simulated circular video SAR. The axis labels and tick marks are omitted and can be found in Fig. 11. (a)–(d) Predetection results for target shadow and energy. The five targets are numbered in yellow text. The correct detections are marked in yellow and false alarms in cyan. The predetection has a high detection probability, but at the cost of a large number of false alarms. (e)–(h) Cumulative interframe-associated tracks until the current frame with a weak logic $\Gamma_{\mathrm{w}} = 3/10$. The associated tracks are marked in different colors, where the white marks represent the reconstructed measurements. The weak-association generates candidate tracks for the following JTA with some false tracks.

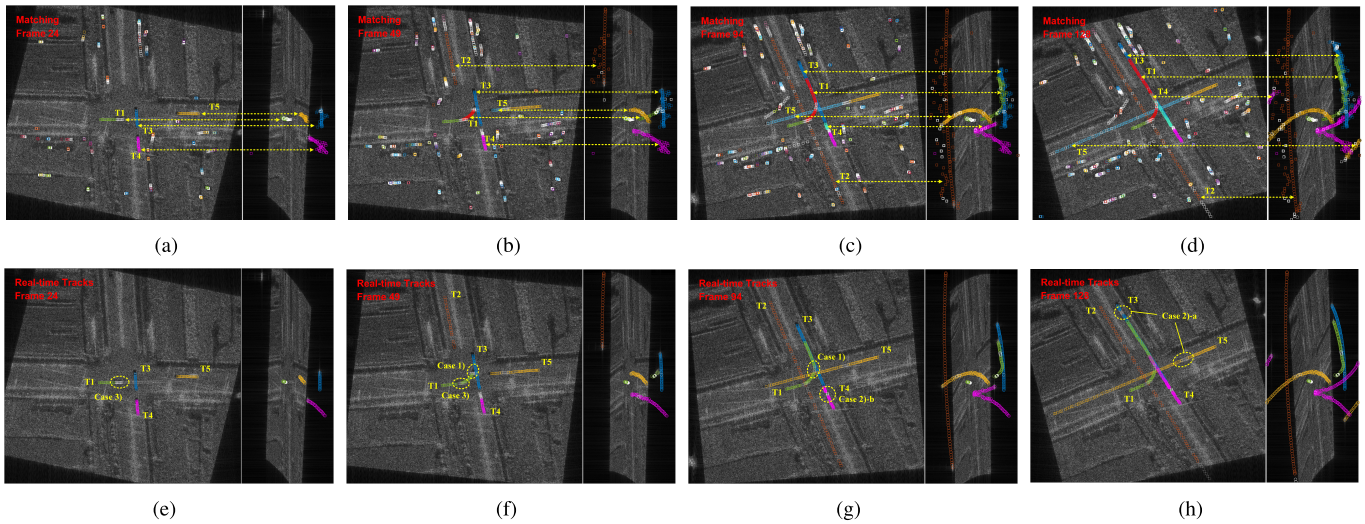

Fig. 13. Matching and tracking results for moving targets in frames 24, 49, 94, and 128. The axis labels and tick marks are omitted and can be found in Fig. 11. (a)–(d) Shadow tracks are projected to the RD domain for matching. The yellow arrows indicate the matching relations for target shadow and energy. (e)–(h) Real-time tracks established by the proposed framework. Several typical cases are indicated by yellow circles. Target T1 is tracked consecutively when facing with undetectable shadow and/or energy by methods in Case 1) and Case 3). Target T2 is tracked effectively by matching the indistinct shadow with its Doppler energy out of the mainlobe clutter. Target T4 is identified and locked for retracking when it comes to a stop according to Case 2)-(b). For targets T3 and T5, the missed shadows are reconstructed effectively with assist of the energy track when facing with the dark background or the bright interference.

RD domain for matching according to (27)–(32), where the ξ_r and ξ_{f_a} are set as 1.3 m and 1240 Hz, respectively. The yellow arrows indicate the matching relations for targets. Fig. 13(e)–(h) depicts the real-time tracks by the proposed framework. The false tracks are eliminated through the JTA and tracks of all five targets are established successfully in the imagery and the RD domain. In the following, each target will be discussed in detail based on the typical cases in Section III-D to show how JTA works.

Target T1 goes straight and turns left at the crossroads and then follows T3 at a constant velocity. The echo energy of T1 in frame 24 is defocused seriously when starts to turn left. The defocused echo energy, as indicated in red dotted rectangles in Fig. 12(a), smears into the shadow area and makes the shadow undetectable, just as stated for Case 3) in Section III-D. Although the prediction is provided by the Kalman filtering and one-step prediction, the track is still interrupted because of the target maneuvering in frame 49, as marked in Fig. 13(f). Since the time interval between the two tracks in the image domain is less than the defined mismatch time ($T_{\mathrm{mis}} = 2$ s), the interrupted tracks are merged by (37) and identified as the same target. In another case,

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TABLE III
STATISTICAL RESULTS FOR THE SIMULATED DATA

Procedure	Correct detection Shadow	& Energy	Correct reconstruction Shadow	& Energy	Missed detection Shadow	& Energy	False alarm Shadow	& Energy
Inter-frame Strong-association	457	380	47	38	143	229	0	0
Inter-frame Weak-association	550	587	46	16	51	44	3224	3939
Joint Track Association (JTA)	545	585	79	28	**23**	**34**	**2**	**7**

The total number of targets to be detected is 647.

the shadow of T1 is covered by the shifted image of T5 in frame 49, which makes the shadow undetectable, as shown in the red dotted rectangle ① in Fig. 12(b). Considering that the energy track is not reformed before the covering and the shadow and energy of T1 have been matched before, this situation corresponds to the Case 1). Therefore, the shadow of T1 is continually tracked by the weak association, as discussed in Section III-D. It is shown that Target T1 can be tracked and reconstructed effectively in the imagery when the shadow is interrupted occasionally. The successful matching between shadow-based track and energy-based track in the following frames provides a more robust tracking for T1 when it moves at a constant velocity later.

Target T2 moves at a high constant velocity during the observation time. The high velocity makes the shadow indistinct and causes many missed detections in the predetection of the image domain. The matching processing between the indistinct shadow and the shifted energy provides a stable tracking for T2. Notice that this framework requires a latency to startup a new target track. Hence T2 is not tracked well in time due to the latency and the missed detections in the imagery when it just enters the scene.

Target T3 moves at a proper velocity during the observation period. It is tracked well even by the strong association. The shadow-based and energy-based tracks also keep matched all the time because of the steady motion state of the target. Some missed measurements of shadow happened at the end of the track due to the darkness of the road, while the echo energy is tracked well in the RD domain, as described in the Case 2)-(a), Section III-D. The missed measurements are recovered via (35) to find out a proper position in the imagery corresponding to the matched echo energy.

Target T4 decelerates and stops for 2.7 s ($>T_{\text{mis}}$), then accelerates and follows with T1 and T3 in a line. In Frame 49, T4 is refocused when it stops [see red dotted rectangle ② in Fig. 12(b)]. The shadow is undetectable while the echo energy keeps detected in a fixed area, just as discussed in Case 2)-(b) in Section III. Target T4 is identified as a go-and-stop target by (36) and the track is locked and retained for re-tracking, which avoids the latency for a new track. When T4 starts moving again, it is retracked immediately by the matched track pair, as marked in Fig. 13(g).

Target T5 slows down before crossing the intersection, then accelerates to resume the velocity, and then moves at a constant velocity. The strong-association track of T5 is intermittent because of the harsh formation condition (see Fig. 11). During the period between frames 24 and 49, the shadow of T5 is covered by its own energy for a while, while the echo energy is tracked well in the RD domain [see Fig. 13(h)]. This situation corresponds to the Case 2)-(a) in Section III. The missed shadows are reconstructed effectively with the assistance of the energy track via (35), as marked by the white rectangles in Fig. 13(h). In addition, the interframe association can also provide a good reconstruction because of the steady motion of the target.

The results show that the five moving targets with typical motion states can be tracked effectively by using our methods. A quantitative analysis with Table III is illustrated for performance evaluation. The number of correct detections, correct reconstructions, missed detections, and false alarms of target shadow and energy are given for the procedure of interframe strong association, interframe weak association, and the JTA processing. The correct detection is defined as the tracking positioning with the distance error less than the priori target size. There are 647 moving targets to be detected in the total 148 frames. In the procedure of interframe strong association, the number of correct detection for shadow and energy are only 457 and 380, respectively, because of the harsh association condition. Although 47 and 38 measurements are reconstructed by the Kalman filter and one-step prediction, 143 and 229 missed detections remain. Fortunately, the strict association also filters out all false alarms, which means that the detection results are with high confidence and can be used as a supplement to the tracking. Comparing with the strong association, the interframe weak association have more correct detections and acceptable missed detection number for both shadow and energy after the reconstruction. However, the large number of false alarms should be eliminated. The JTA takes the advantages of the above two procedures and combines the shadow and energy information to suppress the false alarms. Although the number of correct detections of JTA is slightly less than that of the interframe weak association, the JTA has a better reconstruction ability by the joint processing. The statistics of correct reconstruction and missed detection for each target in imagery are listed in Table IV. In most cases, the target track can be reconstructed well by JTA. The remaining 23 missed detections of shadow mainly come from the beginning of tracking and when T4 comes to stop. In addition, only two false alarms for shadow appears. For target energy in the RD domain, JTA also shows a good performance with 34 missed detections and seven false alarms. The tracks are initiated by the interframe strong association or the matched weak associations between domains, which means that the tracking has a latency of at least M_w frames

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TABLE IV
TRACKING STATISTICS FOR EACH TARGET

Target	Type	Num	Frame No.
T1 1∼148	Reconstruction	25	7, 19∼25, 47∼50, 51∼54, 78, 96, 137∼143
	Missed Detection	4	1, 2, 26, 55
T2 23∼96	Reconstruction	13	46, 51, 55, 62, 64, 65, 69, 71, 73, 75, 90, 91, 96
	Missed Detection	7	23∼29
T3 1∼148	Reconstruction	14	22, 113∼119, 126, 136∼140
	Missed Detection	2	1, 2
T4 1∼148	Reconstruction	9	25∼31, 80, 86
	Missed Detection	6	1, 2, 44∼47
T5 1∼129	Reconstruction	18	34∼40, 77, 94, 97, 100, 105, 114, 120, 126∼129
	Missed Detection	4	1, 2, 41, 42

(here 0.3 s). Therefore, the targets T1, T3, T4, and T5 are tracked effectively after three frames. The target T2 cannot be tracked due to the missed detections when it just enters the scene and is continuously tracked after seven frames. The statistical results show that the proposed framework can effectively decrease the missed detection and false alarm, providing a satisfactory tracking for moving targets with different motion states.

It should be pointed out that the simulation is based on some ideal assumptions. For example, the shadow shape should vary slightly with the change of observation angle because of the height projection of vehicles. Considering the large depression angle in most video SAR imaging, this issue is neglected. The analysis of shadow positioning accuracy can be found in Appendix A. In the following section, an experiment based on the real video data is performed for verification.

B. Results of Real Data

The real SAR video released by the Sandia National Laboratory [28] is used to verify the proposed framework. The radar works in the Ku band and the frame rate is about 3.5 Hz through data overlapping. A total of 386 sequential frames of images and corresponding RD spectra are extracted in the video for our experiment. A target moves along the road with different states such as stopping, accelerating, and turning.

Note that there are some differences between the real data and the simulated data. The radar of the real data does not strictly work in a spotlight mode, but moves in the direction of the moving target and adjusts the observation line of sight (LOS) intermittently. The unknown observation geometry requires us to register the imagery frame by frame additionally. The well-known SIFT [44] and the random sample consensus (RANSAC) algorithm [47] are used for the registration. The accuracy of the solved affine transformation matrix by SIFT have been found to meet the requirement of coordinate alignment in our experiment. In the RD domain, the measurement searching and association are only performed in frames with fixed line of sight, according to (21)–(24). During the adjustment of line of sight, only the established tracks continue to be associated with a nondirectional searching and update, and no new tracks are initiated. In addition, the RD spectra of the real data have been processed by a clutter attenuated Doppler method. This can improve the detection probability of targets greater than a specific velocity, however, at the cost of the detecting failure of the slow targets (see the first few frames). We can still use these RD spectra for verification because the clutter-suppressed spectrum is one of the image types for target energy predetection as discussed in Section III-B. Besides, considering that the Doppler bandwidth is not detailed in the data, only the range constrain by (31) is considered for matching processing in the implementation of JTA algorithm.

The target shadow and energy are first detected with a high detection probability. The predetections are then associated between frames with strong logic value $\Gamma_s = 6/7$ and weak logic value $\Gamma_w = 3/7$, respectively. Next, the matching processing is performed on the weak-associated tracks to suppress the false alarms. Taking frames 90, 204, 300, and 380, for example, the tracking demonstrations are given by Fig. 14. The matching relations are indicated by yellow arrows on the weak-association results as shown in (a)–(d). The cumulative tracks of the target shadow and energy are well established with some false tracks. A complex scene is more likely to produce false tracks than the flat region. After the matching processing, the real-time established tracks are shown in (e)–(h). As seen, the false tracks have been suppressed effectively and the moving target can be tracked timely.

The statistical results are listed in Table V. The number of targets to be detected is 386, which is the same as the frame number. The procedure of interframe strong association establish shadow tracks with high reliability and no false alarms, which can be used as a supplement to the tracking. In the RD domain, there are 36 false alarms for energy even by a strong association due to the frequent adjustment of the line of sight. So, we abandoned the strongly associated tracking results in the RD domain. The interframe weak association has a good performance in the number of missed detection, but it leads to too many false alarms. The JTA algorithm inherits the correct detection number from the weak association and improves the reconstruction ability by the joint processing. The missed 26 shadows and 22 energies are mainly from the first 20 frames when the target is about to start. Only one false track (including 12 and 14 false alarms for shadow and energy) is formed by the wrong matching on the road edge at frame 351.

Two typical cases appear in this data. The shadow-based track is interrupted from frames 64–67 because of the predetection failure for the indistinct shadow. The reconstruction by Kalman also fails to make a successive tracking [see yellow circle in Fig. 14(a)], which corresponds to Case 2)-(a). The JTA reconstructs the track by matching the range cell with the energy successfully [see Fig. 14(e)], which is a degenerate form of (35). From frames 275–278, the shadow-based track is also interrupted like in frame 64. After the reconstruction by matching the energy, the track is still broken because of the inaccurate range center positioning [see yellow circle in Fig. 14(g)]. The split shadow tracks can be judged to come from the same target because the target energy keeps tracking in RD domain. The situation in Fig. 14(g) can also

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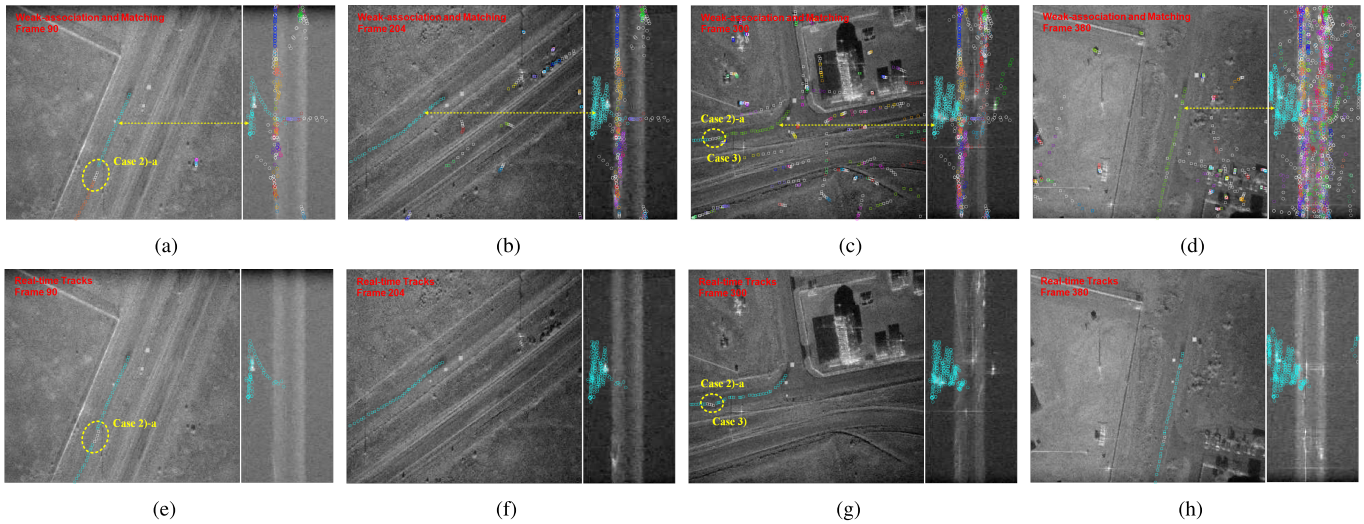

Fig. 14. Demonstrations of moving target tracking by the proposed framework with the video SAR data released by Sandia National Laboratory. (a)–(d) Cumulative weak-associated tracks and matching processing results of frames 90, 204, 300, and 380. The matching relations are indicated by yellow arrows. The track of target shadow is interrupted because of the inaccurate Kalman reconstruction in (a) and (c). The energy keep tracked in RD domain with false tracks. (e)–(h) Established real-time tracks of frames 90, 204, 300, and 380. The missed detections are reconstructed well by the joint processing in (e). The interrupted track is spliced for a consecutive tracking in (g).

TABLE V

STATISTICAL RESULTS FOR THE REAL DATA

Procedure	Correct detection Shadow	& Energy	Correct reconstruction Shadow	& Energy	Missed detection Shadow	& Energy	False alarm Shadow	& Energy
Inter-frame Strong-association	291	214	10	23	85	149	0	36
Inter-frame Weak-association	350	363	7	3	29	20	3960	4135
Joint Track Association (JTA)	349	361	11	3	**26**	**22**	**12**	**14**

The total number of targets to be detected is 386.

be dealt with the Case 3) for a successive tracking according to (37) when the shadow track is interrupted into two tracks. The experimental results demonstrate that the proposed approach works well on the real video SAR data.

A computational test base on the Sandia data is performed on the computer with CPU of Intel Xeon(R) E5-2620. The size of the SAR image is 341×282, and the size of the RD spectrum is 84×282. It takes 1.518 s for per frame by using the proposed framework, where the predetection takes 1.307 s and the JTA algorithm takes 0.209 s. Besides, the time consumed for registration processing is 1.388 s per frame.

C. Comparison

Comparative experiments are carried out with the dynamic programming-based track-before-detect (DP-TBD) method, which has been proved very useful for shadow tracking in video SAR [22]. The DP-TBD method selects the optimal tracking path through the merit function integration between a batch of frames after the target initialization, so it is very suitable for the low-SCNR tracking such as shadow tracking. In our experiments, the target initialization is performed by the background subtraction method as described in [19]. Then the candidate targets are tracked and detected simultaneously over a batch of five frames by integrating the merit function based on the brightness-inverted pixel value. The DP-TBD algorithm declares the presence of moving target shadow if the integrated merit function exceeds the set threshold in the last frame. The step of the batch update is set as 1 and 1 for the simulated and the real data for a better tracking performance, respectively.

TABLE VI

COMPARISON OF THE STATISTICAL TRACKING RESULTS OF SIMULATED DATA AND REAL DATA PROCESSED BY DP-TBD AND JTA

Approach	Simulated data Missed	& False	Real data Missed	& False
DP-TBD	74	52	32	97
JTA	**23**	**2**	**26**	**12**

TABLE VII

COMPARISON OF MISSED DETECTION NUMBER FOR EACH TARGET IN THE SIMULATED DATA

Approach	T1	T2	T3	T4	T5
DP-TBD	10	**2**	**0**	56	6
JTA	**4**	7	2	**6**	**4**

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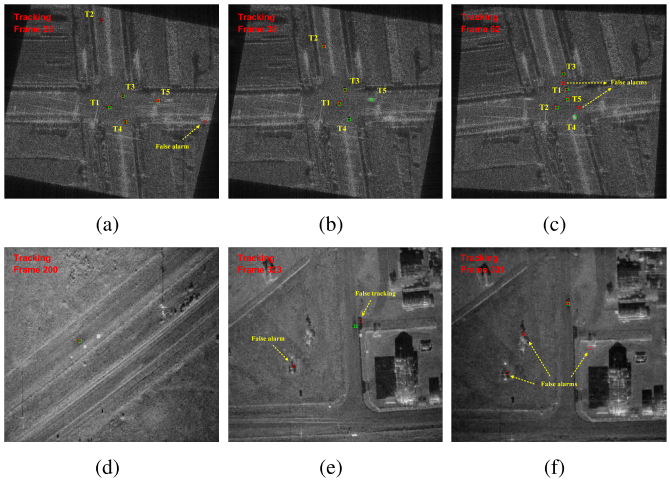

Fig. 15. Comparison of tracking results of the simulated data and real data processed by DP-TBD and JTA. The red rectangle and green rectangle indicate the tracking position by DP-TBD and JTA, respectively. Compared with DP-TBD, the JTA has the advantages of missed detection reconstruction and false alarm suppression. (a)–(c) Tracking results of frames 28, 38, and 62 of the simulated data. (d)–(f) Tracking results of frames 200, 323, and 331 of the real data.

The statistical tracking results of the simulated data and the real data processed by DP-TBD and JTA are compared in Table VI. The proposed JTA has a better tracking performance with less missed detections and false alarms. The tracking results of frames 28, 38, and 62 of the simulated data and frames 200, 323, and 331 of the real data are shown in Fig. 15. The red rectangle and green rectangle indicate the tracking position by DP-TBD and JTA, respectively. For the simulated data, the numbers of missed detections for each target are given by Table VII. The number of missed detections of the targets T2 and T3 processed by DP-TBD is lower than that of JTA, because the indistinct shadows can be tracked more effectively through the merit function integration between frames [see target T2 in Fig. 15(a)]. However, when the target energy smears the shadow area as shown by target T5 in Fig. 15(b), the DP-TBD fails because no pixel can be found to satisfy the constraint of merit function integration. Moreover, it is difficult for DP-TBD to re-initialize the slow or stopping target when starting a new batch of processing, resulting in the tracking failure as shown by target T4 in Fig. 15(b) and (c). In the above two cases, the JTA realizes the reconstruction through the joint processing between domains. Besides, the JTA has a better ability to suppress false alarms, while DP-TBD produces 52 false alarms which mainly comes from the registration error and background subtraction error in the target initialization. For the real data, the number of missed detections and false alarms processed by JTA are both lower than that of DP-TBD. The missed detections processed by DP-TBD and JTA are mainly from the first 20 frames when the target is about to start. In the middle part of the video with homogeneous background and steady adjustment of LOS, the target is tracked well by the two methods as shown by Fig. 15(d). After the target enters the intersection, the prominent objects and frequent adjustment of LOS lead to some false alarms processed by DP-TBD, while the JTA suppress the false alarms effectively by the track association between shadow and energy as shown by Fig. 15(f). Moreover, the improper programming of DP-TBD may generate the false tracking near the target as shown by Fig. 15(e), which makes one missed detection and one false alarm in the statistical results.

The false alarms processed by DP-TBD mainly come from the false target initialization based on background subtraction, while the missed detections mainly come from the smeared shadow and improper programming caused by the nearby interference. Compared with DP-TBD, the proposed framework suppresses the false alarms and reconstructs the missed detections by combining the shadow and energy information between domains. Comparative experiments show that the joint processing improves the tracking performance significantly.

V. Conclusion

The information of a moving target in video SAR can be possibly found by its shadow and echo energy. A joint processing framework has been developed to improve the robustness of moving target tracking significantly. The shadow and echo energy of a moving target are detected in the imagery and the RD spectra using the designed CFAR detectors, respectively. A JTA algorithm has been proposed to associate the predetections. False alarms can be suppressed and the missed detections can be reconstructed by combining the shadow and energy information between domains. Track update and management methods have been proposed to deal with typical cases of target motion states. The proposed framework has been used to process the simulated and the real video SAR data, and the results demonstrate that it can track moving targets robustly.

Appendix A
Positioning Accuracy by Shadow Detection

The shadow is generated by the incomplete integration of background area because of the occlusion of the moving target. There is an error between the position detected by shadow and the real position of the target. The shadow-based positioning accuracy is analyzed here with the radar illumination geometry as shown in Fig. 16, where the radar velocity is $\boldsymbol{v}$ and the grazing angle is ψ.

Assuming that a cube target with height h moves at velocity $\boldsymbol{v}_{\mathrm{t}}$, its footprint area during the integration time is with the shape of ABC'D', where the geometric center M of the footprint area can be used as the accurate position of the target in the current frame. However, considering that the height of the target could also project a shadow area on the ground, the actual shadow area will expand to the polygon area ABEF'G'D'. The detector extracts the actual shadow area and takes its geometric center M' for positioning, which leads to an error along the range direction. The positioning error on the ground plane can be expressed as

$$\Delta p = \left|MM'\right| = \frac{h}{2\tan\psi}. \tag{38}$$

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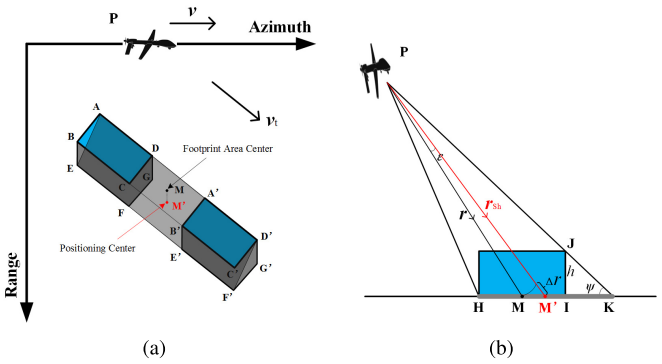

Fig. 16. Illustration of shadow formation. (a) Top view. (b) Section view.

The positioning error mainly depends on the target height and the radar grazing angle. Therefore, the positioning error would not be large for ordinary-height moving targets in a video SAR geometry with high grazing angle. It should be noted that the above analysis is based on the effective formation of the shadow and sufficient imaging resolution.

APPENDIX B
CALCULATION OF MATCHING THRESHOLDS ξ_r AND ξ_{f_a}

The JTA algorithm associates the shadow tracks with the energy tracks for a robust tracking. The matching processing by formulas (31) and (32) give the rules to eliminate false alarms. The matching thresholds ξ_r and ξ_{f_a} are determined by the inconsistency between the shadow positioning and energy positioning.

As shown in Fig. 16(b), the range vector from radar to the target real position and detected shadow position are given by $\boldsymbol{r}$ and $\boldsymbol{r}_{\rm sh}$, respectively. Assuming that the range positioning in the RD spectrum is accurate after the range walk correction and the measurement error of geometric center caused by image processing are not considered temporarily, the range inconsistency between the shadow projection and the direct readout in RD spectrum can be expressed as

$$\Delta r = \|\boldsymbol{r}_{\rm sh}\|_2 - \|\boldsymbol{r}\|_2 \approx \left|MM'\right| \cos\psi = \frac{h\cos\psi}{2\tan\psi}. \tag{39}$$

Then, ξ_r is defined as follows:

$$\xi_r = \kappa_r \Delta r = \kappa_r \frac{h\cos\psi}{2\tan\psi} \tag{40}$$

where κ_r denotes the range tolerance factor slightly greater than 0 to against the measurement error.

Similarly, ξ_{f_a} is determined by the Doppler inconsistency between the shadow projection and the direct readout in RD spectrum. The actual Doppler and the shadow projected Doppler can be expressed as

$$f_a = \frac{2}{\lambda}(\boldsymbol{v} - \boldsymbol{v}_{\rm t}) \cdot \frac{\boldsymbol{r}}{\|\boldsymbol{r}\|_2} \tag{41}$$

and

$$f_{a,pro} = \frac{2}{\lambda}(\boldsymbol{v} - \hat{\boldsymbol{v}}_{\rm Sh}) \cdot \frac{\boldsymbol{r}_{\rm Sh}}{\|\boldsymbol{r}_{\rm Sh}\|_2} = \frac{2}{\lambda}(\boldsymbol{v} - \boldsymbol{v}_{\rm t} - \hat{\boldsymbol{v}}_{\rm e}) \cdot \frac{\boldsymbol{r}_{\rm Sh}}{\|\boldsymbol{r}_{\rm Sh}\|_2} \tag{42}$$

respectively, where $\hat{\boldsymbol{v}}_{\rm e}$ denotes the error of the estimated velocity by shadow, which is mainly determined by the SAR image quality and measurement error and can be chosen on an experimental basis. Then, the Doppler inconsistency can be expressed by

$$\begin{aligned}\Delta f_a &= \left\| f_{a,pro} - f_a \right\|_1 \\ &= \left\| \frac{2}{\lambda}(\boldsymbol{v} - \boldsymbol{v}_{\rm t}) \cdot \left(\frac{\boldsymbol{r}_{\rm Sh}}{\|\boldsymbol{r}_{\rm Sh}\|_2} - \frac{\boldsymbol{r}}{\|\boldsymbol{r}\|_2} \right) - \frac{2}{\lambda}\hat{\boldsymbol{v}}_{\rm e} \cdot \frac{\boldsymbol{r}_{\rm Sh}}{\|\boldsymbol{r}_{\rm Sh}\|_2} \right\|_1 \\ &\le \frac{2}{\lambda}(\|\boldsymbol{v}\|_2 + \|\boldsymbol{v}_{\rm tmax}\|_2)\varepsilon + \frac{2}{\lambda}\|\hat{\boldsymbol{v}}_{\rm e}\|_2 \cos\psi \end{aligned} \tag{43}$$

where $\boldsymbol{v}_{\rm tmax}$ denotes the maximum possible velocity of the moving target, and ε denotes the small angle between $\boldsymbol{r}$ and $\boldsymbol{r}_{\rm Sh}$ that can be approximated as

$$\varepsilon \approx \frac{\left|MM'\right| \sin\psi}{\|\boldsymbol{r}\|_2} = \frac{h\cos\psi}{2\|\boldsymbol{r}\|_2}. \tag{44}$$

The thresholds ξ_{f_a} is defined as follows:

$$\xi_{fa} = \kappa_{f_a}\left[\frac{2}{\lambda}(\|\boldsymbol{v}\|_2 + \|\boldsymbol{v}_{\rm tmax}\|_2)\varepsilon + \frac{2}{\lambda}\|\hat{\boldsymbol{v}}_{\rm e}\|_2 \cos\psi\right] \tag{45}$$

where κ_r denotes the Doppler tolerance factor slightly greater than 1 against the measurement error of Doppler center in the RD spectrum.

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